Harsh Desai

Ann Arbor, MI | harshdes@umich.edu | +1-734-548-1080 | LinkedIn | Portfolio

Mechanical and Space Systems Engineer experienced in CAD/CAE, electromechanical mechanisms, scientific instrumentation, vacuum/HV test stands, subsystem integration, and design validation.

EDUCATION

University of Michigan — **Ann Arbor, MI**
Master of Engineering, Space Systems Engineering | GPA: 3.79/4.0 — Aug 2024-Dec 2025

Vellore Institute of Technology — **Vellore, India**
Bachelor of Technology, Mechanical Engineering | GPA: 7.78/10 — Jul 2019-Jul 2023

TECHNICAL SKILLS

Mechanical System Design & CAE: SolidWorks, Fusion 360, Autodesk Inventor, AutoCAD, ANSYS Mechanical/Fluent, SpaceClaim; conceptual design, 3D part/assembly modeling, structural and thermal-fluid analysis

Prototype & Test: mechanism integration, component selection and procurement, assembly, troubleshooting, test procedures, design verification, FMEA, vacuum-chamber operation, high-voltage test stands

Instrumentation & Automation: plasma diagnostics, sensors and detector systems, electrostatic ion optics, Python, PyVISA/PyMeasure, LabVIEW (CLAD preparation), MATLAB, Keithley DAQs/SMUs, TDK Lambda power supplies

Documentation & Analysis: Microsoft Excel, Word, PowerPoint; test procedures, FMEA, requirements and verification documentation

ENGINEERING EXPERIENCE

Climate and Space Sciences and Engineering, University of Michigan — **Ann Arbor, MI**
Research Assistant, LVACCS Experimental Hardware & Test Automation — Feb 2026-Present

- Integrated and validated a 1.3 kV plasma-source test stand spanning vacuum hardware, a custom discharge interface, power supplies, Keithley acquisition, and synchronized Python/PyVISA control; reduced manual steps from 15 to 5
- Developed a discharge-box FMEA, ignition and acquisition procedures, and post-run checks; achieved 98.6% synchronized data capture and improved repeatability across hardware test states
- Operated Langmuir and emissive probes plus an RPA assembly during vacuum-chamber testing, performing LP/EP measurements and synchronized RPA retarding-voltage/collector-current acquisition while resolving mechanical/electrical interface issues

Space Physics Research Lab, University of Michigan — **Ann Arbor, MI**
Summer Intern, TestBedz Spacecraft Qualification Platform — May 2025-Jul 2025

- Built a requirements-to-test workflow for thermal-vacuum, vibration, and EMI qualification, matching hardware to facility limits across six test types
- Supported subsystem design reviews and technical presentations by translating operating envelopes and acceptance criteria into compatibility checks and traceable test plans for repeatable handoff to test facilities

Solar and Heliospheric Research Group, University of Michigan — **Ann Arbor, MI**
Graduate Research Assistant, Solid-State Detector Readout — Jan 2025-Dec 2025

- Derived ADC requirements from detector energy resolution, noise, signal amplitude, and timing; evaluated a 14-bit, 125 MSPS ADC integration with a Zynq-7000 development platform
- Verified MATLAB HDL Coder models against Vivado co-simulation and diagnosed 125 MHz post-synthesis timing constraints, documenting the design limits and next integration steps

SELECTED MECHANICAL & INSTRUMENTATION PROJECTS

CanSat 2022 (NASA/AAS) | 7th worldwide of 42 teams — Aug 2021-Jul 2022

- Designed and built a compact 10 m tether-deployment and camera-stabilization assembly using a DC motor, worm gear, bearings, an ABS printed structure, and a dual-servo gimbal for controlled descent and fixed 45-degree imaging
- Iterated the deployment architecture from spur to worm gearing for one-way load transfer, then worked across the mechanical and electronics team to integrate the mechanism with payload electronics and verify release and imaging functions through subsystem tests and design reviews

Spaceport America Cup / IREC | 23rd globally, 5th Asia-Pacific — Apr 2020-Jul 2021

- Developed an in-house trajectory model for a Mach 0.9, 10,000 ft rocket, generating max-Q pressure and drag inputs in ANSYS Fluent and cross-checking the flight prediction in OpenRocket

Particle Detector & Electrostatic Analyzer Calibration — Jul 2025-Dec 2025

- Calibrated a channel electron multiplier (CEM) signal chain by configuring the ion source, CEM, charge-sensitive preamplifier, pulse shaper, and MCA; charge-calibrated MCA channels with an RC injector and analyzed pulse-height spectra to extract centroid, FWHM, count rate, and effective gain versus detector bias
- Modeled a quadrispherical electrostatic analyzer in SIMION across an 84-point energy-angle grid, quantifying pass-energy shift, energy resolution, angular acceptance, and fringe-field sensitivity